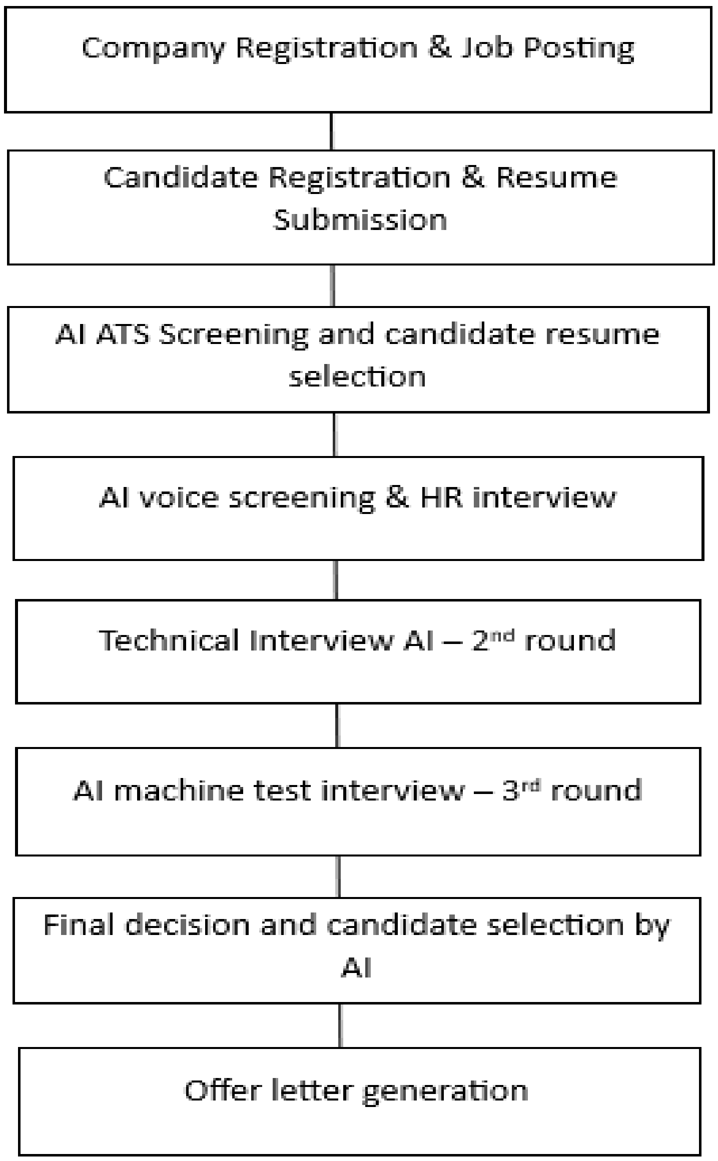
Zecpath – AI-Powered Job Portal & Automated Hiring Assistant

Zecpath is an AI-powered job portal and hiring platform which helps in faster and efficient hiring process for companies across industries and job roles. It can reduce manual resume screening using AI ATS and schedule interviews automatically based on roles and availability.

1. Hiring Lifecycle Flow Chart



2. AI Engines and their Responsibilities

- NLP (Natural Language Processing):
  - Understands and processes human language for chatbots, interaction, and voice assisants.

- Custom ML (Machine Learning) Models:
  - Trained on specific datasets for automation according to the requirement of company instead of a general one.

- Resume Parser AI:
  - Extracts and organizes candidate information such as skills, projects, experience and more.

- Resume Screening AI:
  - Analyzes resumes and filters candidates based on JDs and keywords.

- Behavioral Analysis:
  - Evaluates candidate’s behavior5, tone, confidence and interaction5.

- TTS (Text-to-Speech):
  - Converts written text into speech as human-form for audio output.

- STT (Speech-to-Text):
  - Converts an audio spoken into text for analysis and processing.

- Voice Personalization AI engine:
  - Adjusts voice interaction based on language, tone or gender-based preferences for the candidate.

3. Zecpath AI Responsibilities Overview

The platform helps to automate and enhance the recruitment and the interview process. AI (Artificial Intelligence) plays a major role in improving the process with faster and efficient user experience. In the project, several AI modules have been covered and plays an important factor in each step of the recruitment process. Among them, the first one is NLP. Natural Language Processing, it understands human interaction or the language and hence enables text analysis, communication, resume parsing and improves the gap between the users and the system.

The next one is the use of some custom-made Machine Learning models which are trained for the specific tasks based on the requirements and workflows of the platform. Different companies may require different recruitment strategies or candidate evaluation criteria or the scoring during the tests in the hiring process. In addition, the platform consists of the ATS (Application Tracking System) AI which helps manage candidate application, organize and rank them based on the skills, job profile and the hiring procedure. This way it could analyze and filter or shortlist the candidates by processing resumes and job-related text data. Thus, candidate could receive auto-generated selection or rejection mail based on the evaluation results. Selected candidates are also provided with AI-powered reminder calls and notifications regarding the scheduled interview time, assessment or any updates.

The platform also includes a Malpractice and Integrity Detection system to ensure fairness during online interviews and assessments. The system monitors activities such as repeated tab switching, frequent looking away from the screen, detection of external voices, and possible mobile phone usage during the interview process. If any suspicious activity is identified, the platform can display warning prompts. This helps maintain the integrity and reliability of the recruitment process.

The behavioral analysis AI evaluates candidate behavior during the AI HR interviews, The AI HR is made capable of analyzing factors such as communication, confidence, tone of speech, interaction patterns and the language accent they prefer. This could improve the effectiveness of the hiring platform by providing behavioral factors of the candidate along with candidate's technical qualification and resume screening.

The platform integrates both TTS and STT to improve user interaction throughout the hiring and interview process . The Text-to-Speech converts written text into speech as human-form for audio output. This helps the AI HR assistant to communicate with candidates throughout the interview like a dynamic approach. When a candidate answers certain questions, the AI hr could ask questions based on the answers of the candidate provided by them in the previous question. This is known as the dynamic follow-up. Also, the candidate could ask questions back to the AI hr regarding anything about the company or the job. The Speech-to-Text engine is responsible for converting spoken language into text in real time, enabling the system to process candidate responses during voice-based or video interviews.

After the successful completion of the machine test rounds, aptitude assessments, and overall candidate evaluation, the platform proceeds with further recruitment processes. Based on the candidate's overall performance throughout the recruitment process, the system further includes an AI-powered Salary Negotiator tool that assists in determining suitable salary discussions according to candidate skills, performance scores, and company requirements. Finally, after successful completion of all recruitment stages, the platform automatically generates the final offer letter and related onboarding documents, thus these AI-driven components work together to reduce manual effort, improve candidate selection accuracy, and enhance the overall efficiency and intelligence of the hiring process within the platform.

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